Def. 4.1.1 An associative algebra over $\mathbb{C}$ is a vector space A over $\mathbb{C}$ together with a multiplication map $A \times A \longrightarrow A$, $(x,y) \longmapsto xy$ which is bilinear and associative. We say A is unital (or A has a unit) if there exists an element $e \in A$ such that $ex = xe = x$ for all $x \in A$.

Remark The unit element of A, if exists, is usually denoted by 1.

Examples

* $End(V)$ for any $\mathbb{C}$-vector space V.
* $Mat_{n\times n}(\mathbb{C})$ with usual matrix operations.
* The subspace of upper triangular matrices in $Mat_{n\times n}(\mathbb{C})$
* The subspace of strictly upper triangular matrices in $Mat_{n\times n}(\mathbb{C})$

  (is this algebra unital?)

Example Let G be a finite group. Then $A[G]$ (sometimes denoted $\mathbb{C}[G]$) is defined as follows: $A[G] := \{\varphi : G \longrightarrow \mathbb{C}\}$. Note that $A[G]$ is a $\mathbb{C}$-vector space: $(\varphi_1 + \varphi_2)(g) := \varphi_1(g) + \varphi_2(g)$; $(c\cdot\varphi)(g) := c\cdot(\varphi(g))$ $c \in \mathbb{C}$

* For $g \in G$ we set $\delta_g(h) := \begin{cases} 1 & h = g \\ 0 & h \neq g \end{cases}$. Then $\{\delta_g\}_{g\in G}$ is a basis for the $\mathbb{C}$-vector space $A[G]$.
* We define $\delta_{g_1} \cdot \delta_{g_2} := \delta_{g_1 \cdot g_2}$. We extend this multiplication to a bilinear map $A[G] \times A[G] \longrightarrow A[G]$.

Thus: $\left(\sum_{g\in G} x(g)\cdot\delta_g\right)\cdot\left(\sum_{h\in G} y(h)\cdot\delta_h\right) := \sum_{g,h\in G} x(g)\,y(h)\,\delta_{gh}$

Remark The book considers a slightly more general setting: finite-support functions on any group

Example $G = \mathbb{Z}_2 = \{\bar{0}, \bar{1}\}$. Then $A[G] = \text{Span}\{\delta_{\bar{0}}, \delta_{\bar{1}}\}$.

$$(x\delta_{\bar{0}} + y\delta_{\bar{1}}) \cdot (x'\delta_{\bar{0}} + y'\delta_{\bar{1}}) = (xx' + yy')\delta_{\bar{0}} + (xy' + x'y)\delta_{\bar{1}}.$$

Remark: $A[G]$ is always unital with unit δ_e ($e \in G$: neutral element of G)

Def 4.1.2 Let A be an associative algebra. A representation of A is a pair (ρ, V) where:

* V is a vector space over $\mathbb{C}$.
* $\rho : A \longrightarrow \text{End}(V)$ is a homomorphism of associative algebras. That is,
  - ρ is $\mathbb{C}$-linear.
  - $\rho(ab) = \rho(a)\rho(b) \quad \forall a, b \in A$
  - If A is unital with unit $1 \in A$, then $\rho(1) = I$.

Remark Given a representation (ρ, V) of an associative algebra A, we sometimes call V an A-module.

Def. 4.1.2 (Continued) Let V be an A-module. Then a subspace $U \subseteq V$ is called an A-submodule (or just submodule) of V iff $\rho(a)U \subseteq U$ for all $a \in A$.

Example $A = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} : a, b, c \in \mathbb{C} \right\} \subseteq \text{Mat}_{2\times 2}(\mathbb{C})$. Let $V = \mathbb{C}^2$ be the standard module (i.e. elements of A act on V by matrix multiplication).

Then $U := \text{Span}_{\mathbb{C}}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} t \\ 0 \end{bmatrix} : t \in \mathbb{C} \right\}$ is an A-submodule of V.

Some Constructions with A-modules.

* Let (ρ, V) and (σ, W) be two A-modules. Then $V \oplus W$ has a canonical A-module structure: We set: $\eta : A \longrightarrow \mathrm{End}(V \oplus W)$

$$\eta(a)(v \oplus w) = \rho(a)v \oplus \sigma(a)w \qquad (v \in V,\ w \in W,\ a \in A).$$

Then $(\eta, V \oplus W)$ is an A-module.

* Let V be a A-module and let $U \subseteq V$ be an A-submodule.

Then the quotient $V/U := \{ U + v : v \in V \}$ is an A-module:

We define $\bar{\rho} : A \longrightarrow \mathrm{End}(V/U)$ $\quad \bar{\rho}(a)(U+v) = U + \rho(a)v$

* $\bar{\rho}$ is well-defined because:

$$\forall v, v' \in V: \quad U + v = U + v' \Rightarrow v - v' \in U \Rightarrow \rho(a)(v - v') \in U$$

$$\Rightarrow \rho(a)v - \rho(a)v' \in U$$

$$\Rightarrow U + \rho(a)v = U + \rho(a)v'.$$

<u>Remark</u> Let A be an associative algebra. A subspace $I \subseteq A$ is called a (two-sided) ideal of A iff $aI + Ia \subseteq I$ for all $a \in A$.

<u>Example</u> $A := \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} : a, b, c \in \mathbb{C} \right\} \subseteq \mathrm{Mat}_{2\times 2}(\mathbb{C})$.

Then $I := \left\{ \begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} : t \in \mathbb{C} \right\}$ is an ideal of A.

<u>Exercise</u> Prove that the only ideals of $A := \mathrm{Mat}_{n\times n}(\mathbb{C})$ are $\{0\}$ and A.

Lemma Let ρ be an associative algebra and let (ρ, V) be an A-module. Then

$\ker(\rho) := \{a \in A : \rho(a)V = \{0\}\}$ is an ideal of A.

Def. 4.1.3 Let (ρ, V) and (σ, W) be two A-modules where A is an assoc. alg.

We define $\operatorname{Hom}_A(V, W)$ to be the $\mathbb{C}$-vectorspace of linear transformations

$T : V \longrightarrow W$ which satisfy $T\rho(a) = \sigma(a)T$ for all $a \in A$.

Elements of $\operatorname{Hom}_A(V,W)$ are called intertwining operators

We say V and W are isomorphic (or equivalent) A-modules iff there exists an invertible $T \in \operatorname{Hom}_A(V,W)$. Note that in this case $T^{-1} \in \operatorname{Hom}_A(W,V)$.

Example $A = \mathbb{C}[x]$, V any vector space, $T \in \operatorname{End}(V)$. Then we obtain an A-module $\rho_T : A \longrightarrow \operatorname{End}(V)$ $\quad \rho(f(x)) := f(T)$

for every $f(x) \in \mathbb{C}[x] = A$.

$\ker(A)$ = ideal generated by minimal polynomial of T

(recall that $\mathbb{C}[x]$ is a PID).

If $\dim(V) < \infty$, the problem of classifying representations of $\mathbb{C}[x]$ up to equivalence can be reduced to the Jordan canonical form of T.

Example Let G be any group and let $\rho : G \longrightarrow GL(V)$ be a group homomorphism. $GL(V) = \{T : V \longrightarrow V, \mathbb{C}\text{-linear and invertible}\}$. Then V has a canonical $A[G]$-module structure: we define

$\tilde{\rho} : A[G] \longrightarrow \operatorname{End}(V)$ $\quad \tilde{\rho}\left(\sum x(g)\delta_g\right) := \sum x(g)\rho(g)$.